

OSCILLATIONS

- Periodic motion:- Repeats itself after regular interval of time.

e.g. uniform circular motion or orbital motion. An insect climbing wall and falling down repeatedly. motion of a bouncing ball (no loss of energy) motion of a pendulum.

- Oscillatory motion / Harmonic motion

- It is a part of periodic motion.
- To and fro (back and forth or up and down) about fixed point after regular interval of time.
- Fixed point about which body oscillates is called mean position or equilibrium position.
 $F_{net}, a_{net} = 0$

e.g. motion of pendulum
spring mass system.

- All oscillatory motion (with no loss of energy) are periodic but all periodic motion are not oscillatory.

- Condition for any oscillatory / harmonic motion

$$F = -Kx^n \quad (n = 1, 3, 5, 7, \dots \text{any odd integer})$$

Restoring force $\rightarrow$ Constant

* (Always acts towards mean position)

- Simple Harmonic Motion

$$F = -Kx \quad (n = 1)$$

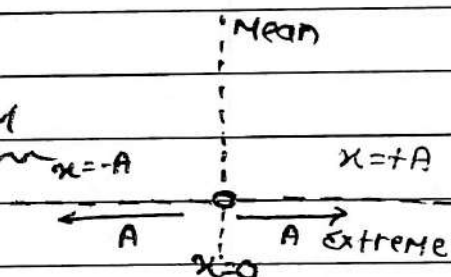
$\downarrow$
displacement from mean position

$\rightarrow$ F depends on x ($\uparrow x = \uparrow F$)

- Basic Terms Related to S.H.M

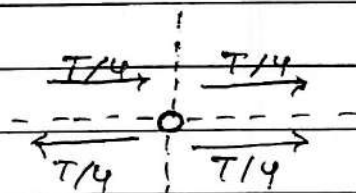
<1> Amplitude (A)

- Extreme position covered by mass



<2> Time Period (T)

• Time for 1 complete oscillation.



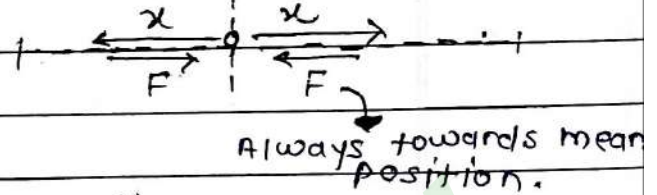
<3> frequency and Angular frequency

$$f = T^{-1} \text{ or } \frac{1}{T} \quad (\text{in } s^{-1} \text{ or hertz})$$

$$[\omega = 2\pi f = \frac{2\pi}{T}] \quad (\text{in rad/s})$$

(4) Restoring force

$$[F = -kx]$$



$$[k = m\omega^2] \quad (m = \text{mass of particle})$$

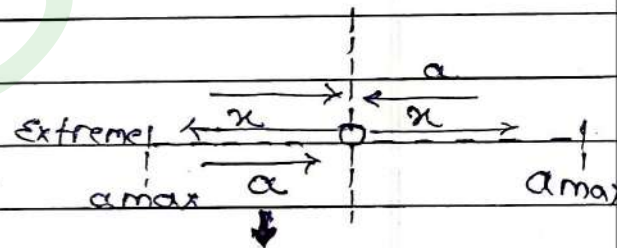
$$[F = -m\omega^2 x]$$

- ★ SHM is a type of oscillation in which acceleration of a particle is directly proportional to displacement from mean position and opp. to it.

(5) Acceleration (a)

$$\frac{\vec{a}}{m} = \frac{\vec{F}}{m} = -\frac{kx}{m} = -\frac{m\omega^2 x}{m}$$

$$[a = -\omega^2 x]$$

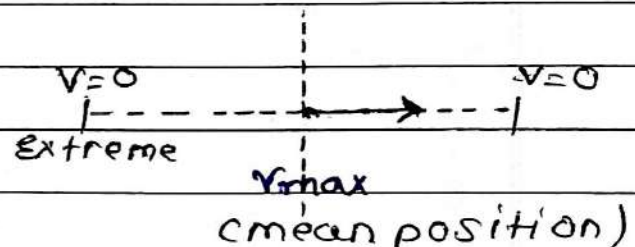


(always towards mean position)
($x=0, a=0$)

$$[|a_{\max}| = \omega^2 A]$$

(6) Velocity (v)

$$[v^2 = \omega^2 (A^2 - x^2)]$$



$$[v = \pm \omega \sqrt{A^2 - x^2}]$$

★ At mean position, velocity is maximum.

$$[|v_{\max}| = \omega A]$$

>> The distance covered by a particle under-going SHM in one time period is
 (Amplitude = A) = $4A$.

Que-1 A particle executes linear SHM with amplitude of 3cm. When the particle is at 2cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in second is?

Sol $A = 3\text{cm}$ $T = \frac{2\pi}{\omega} = \frac{2\pi \times 2}{\sqrt{5}} = \frac{4\pi}{\sqrt{5}}$

$x = 2\text{cm}$

$|v| = |a|$

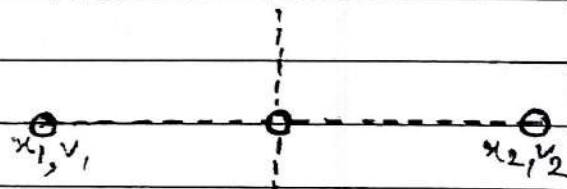
$\omega \sqrt{A^2 - x^2} = \omega^2 x$

$\sqrt{9-4} = \omega \times 2$

$\omega = \frac{\sqrt{5}}{2}$

>> A particle is executing SHM along a straight line. Its velocities at distance x_1 and x_2 from the mean position are v_1 and v_2 , then its time is -

$$T = 2\pi \sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$



>> Differential Equation for S.H.M

$$a = -\omega^2 x$$

$$a = \frac{dv}{dt} = \frac{d\left(\frac{dx}{dt}\right)}{dt} = \frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$\frac{d^2x}{dt^2} + \omega^2 x = 0$	→ Condition for S.H.M ($x \propto a$)
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>> Equation for S.H.M

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

↓ Soln

$x = A \sin(\omega t + \phi)$	(if $t=0$, $\phi \rightarrow$ initial Phase)
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↓ disp. from mean position ↓ Amplitude ↓ Phase

$0 \leq \phi < 2\pi$

$$x = A \cos(\omega t + \phi)$$

$$x = A \sin \omega t + B \cos \omega t$$

$$x = A + B \sin \omega t$$

} SHM

>> Equation of displacement (x), Velocity (v) and acceleration (a) in term of time (t) for SHM whose initial position is as shown.

$$x = A \sin (\omega t + \phi)$$

$$0 = A \sin \phi$$

$$\sin \phi = 0$$

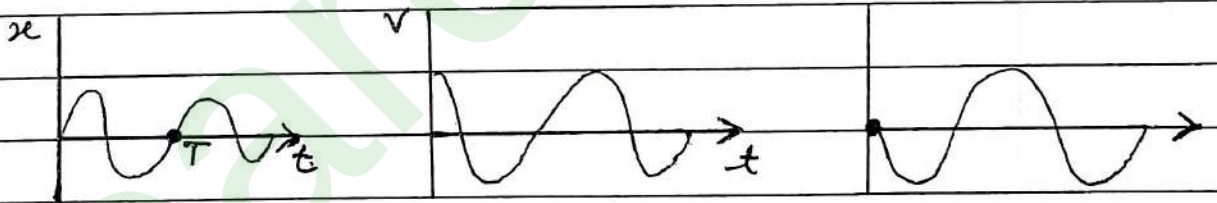
$$\phi = 0, \pi, (v = +ve)$$

$$\star [x = A \sin \omega t]$$

$$[v = \frac{dx}{dt} = \omega A \cos \omega t]$$

$$[a = -\omega^2 A \sin \omega t]$$

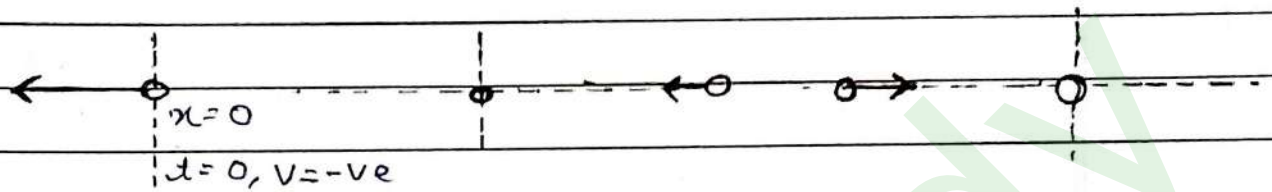
Graph



Phase Difference

$$\begin{aligned} x &= A \sin \omega t \quad \leftarrow \pi/2 \\ v &= \omega A \cos \omega t \quad \leftarrow \pi/2 \\ a &= -\omega^2 A \sin \omega t \end{aligned}$$

>> Average velocity of a particle executing SHM in one complete vibration is zero (disp = 0).



$x = -A \sin \omega t$	$x = A \cos \omega t$	$x = -A \cos \omega t$
$v = -\omega A \cos \omega t$	$v = -\omega A \sin \omega t$	$v = \omega A \sin \omega t$
$a = \omega^2 A \sin \omega t$	$a = -\omega^2 A \cos \omega t$	$a = \omega^2 A \cos \omega t$

S.H.M as projection of uniform Circular motion

- uniform circular motion is not a S.H.M.
- Projection on two diameters is doing S.H.M but particle does not.

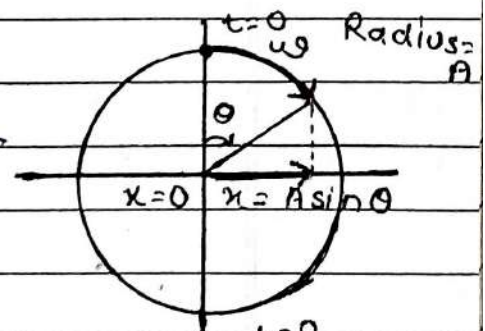
► Projection on Horizontal diameter

Particle - circular motion

Angular velocity $\rightarrow \omega$

$$x = A \sin \theta$$

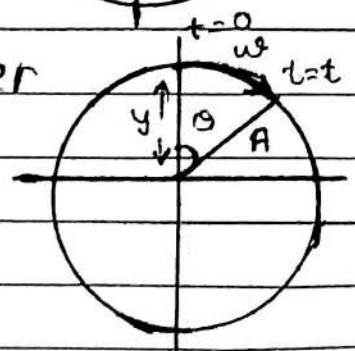
$$x = A \sin \omega t \quad (\theta = \omega t)$$



► Projection on vertical diameter

$$y = A \cos \theta$$

$$y = A \cos \omega t$$



⇒ Time taken by the particle to go from its mean position to half of its Amplitude is $\left(t = \frac{T}{12} \right)$

Energy in S.H.M

► Kinetic Energy

$$KE = \frac{1}{2} mv^2 = \frac{1}{2} m\omega^2 (A^2 - x^2) \quad [v^2 = \omega^2 (A^2 - x^2)]$$

$$KE = \frac{1}{2} k(A^2 - x^2) \quad [k = m\omega^2]$$

Extreme

$$x = \pm A$$

$$KE = 0$$

(minimum)

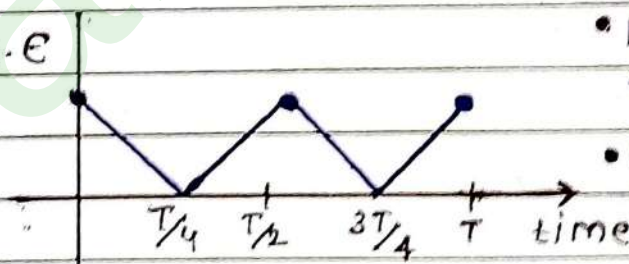
Mean position

$$x = 0$$

$$KE = \frac{1}{2} kA^2$$

(maximum)

GRAPH K.E



• KE complete two cycle in one oscillation

• Frequency of KE is double SHM.
($f' = 2f$)

► $k \cdot E$ v/s time

$$(v = A\omega \cos \omega t)$$

$$KE = \frac{1}{2} mv^2$$

$$= \frac{1}{2} (m A^2 \omega^2) \cos^2 \omega t$$

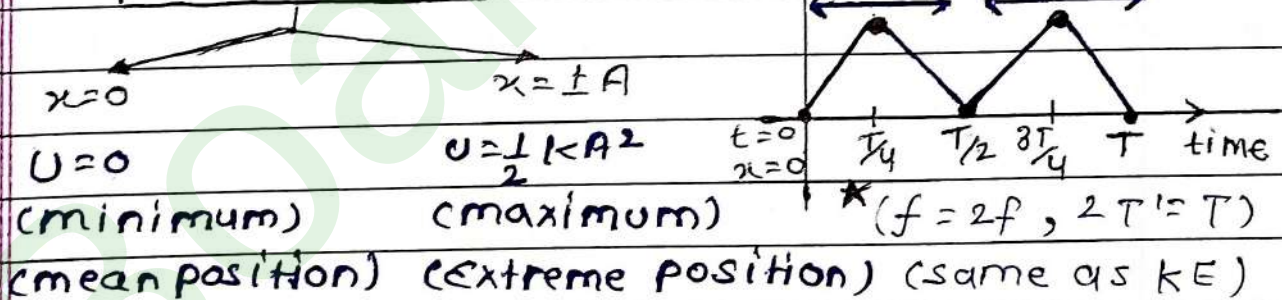
$$= \frac{1}{2} k A^2 \cos^2 \omega t$$

$$\boxed{KE = k_{\max} \cos^2 \omega t} \quad (k_{\max} \rightarrow \text{max. kinetic Energy})$$

cos is in square so its Graph can never be negative.

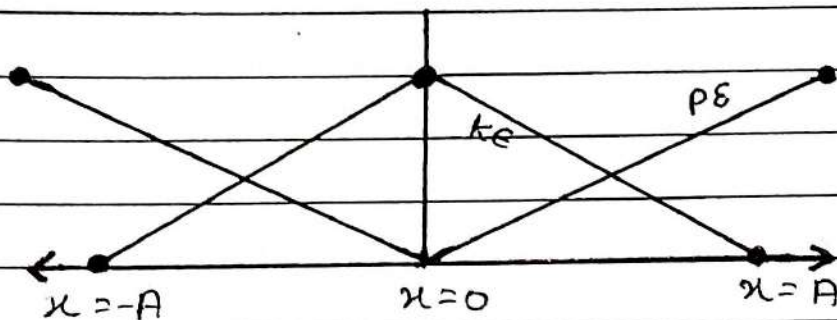
► Potential Energy

$$\boxed{U = \frac{1}{2} kx^2} \quad \boxed{= \frac{1}{2} k A^2 \sin^2 \omega t = U_{\max} \sin^2 \omega t}$$



$$\boxed{U_{\max} = KE_{\max} = \frac{1}{2} k A^2}$$

► Graph of KE & PE v/s x



>> Displacement b/w max. PE position and max. KE position for a particle executing SHM is $\Rightarrow \pm a$ (amplitude)

► Total Mechanical Energy in SHM: * over a period of oscillation,

$$T.E = KE + U = \frac{1}{2} kA^2 - \frac{1}{2} kx^2 + \frac{1}{2} kx^2$$

$$\boxed{T.E = \frac{1}{2} kA^2} = \text{CONSTANT}$$

Average KE =

$$\text{Avg. PE} = \frac{1}{4} m\omega^2 A^2 \quad \text{OR}$$

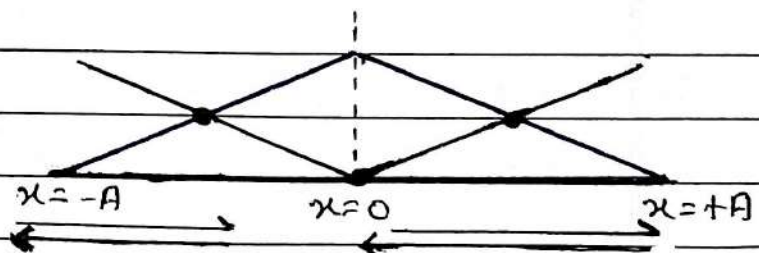
$$= \frac{1}{4} kA^2$$

$$TE = k_{\text{max}} = U_{\text{max}} = \frac{1}{2} kA^2$$

(mean) (extreme)

>> In SHM during one complete period, $PE = KE$ for 4 times

$$\left(\text{at } x = \pm \frac{A}{\sqrt{2}} \right)$$



► Steps to Find Time Period of any S.H.M

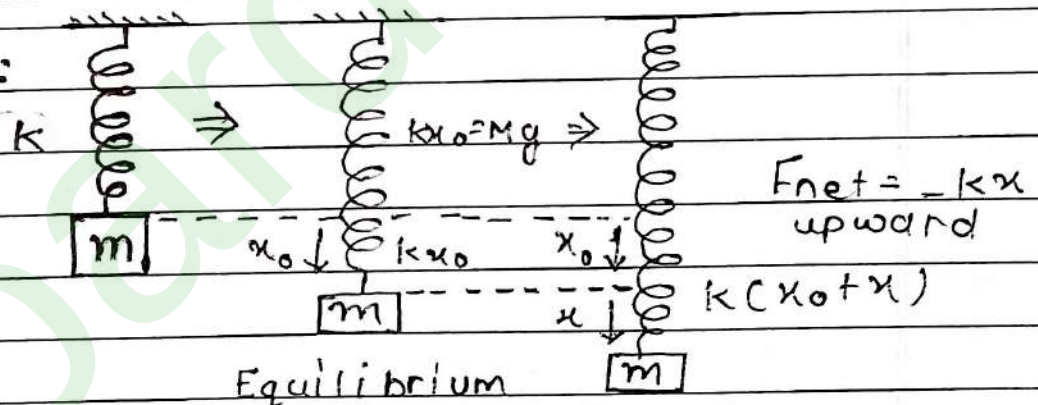
1. Find the equilibrium position where $F_{\text{net}} = 0$ on body.
2. Displace the body slightly (x), from its position & Find net force on body now. Express net force in form of $F = -kx$

3. $k = m\omega^2$

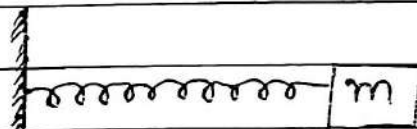
$$\omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}} = \frac{2\pi}{T} \Rightarrow T = 2\pi \sqrt{\frac{m}{k}}$$

► Spring Mass System / Oscillator

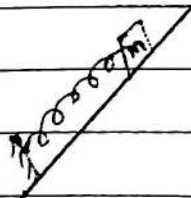
Case I:



Case II:



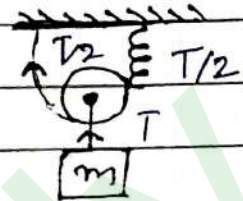
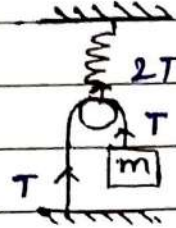
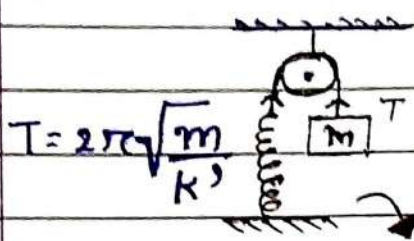
Case - III:



For case I, II & III

$$T = 2\pi \sqrt{\frac{m}{k}}$$

Case IV :- mass is attached with string and string with spring.



$$K' = \left[\frac{T_m}{T_s} \right]^2 K$$

$$K' = \left(\frac{T}{T} \right)^2 K$$

$$K' = \left(\frac{T}{2T} \right)^2 K$$

$$K' = \left(\frac{T}{T/2} \right)^2 K$$

$$T_m = T_{\text{in mass}}$$

$$K' = K$$

$$K' = \frac{K}{4}$$

$$K' = 4K$$

$$T_s = T_{\text{in string}}$$

$$T = 2\pi\sqrt{\frac{m}{K}}$$

$$T$$

$$K' = \text{effective } k$$

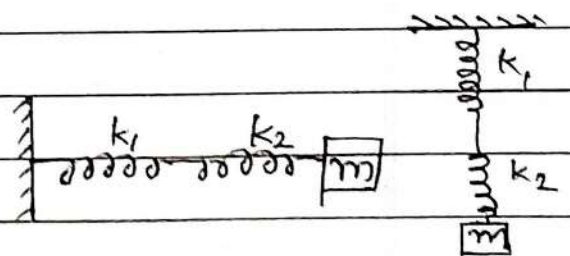
$$T = 2\pi\sqrt{\frac{4m}{K}}$$

$$T = 2\pi\sqrt{\frac{m}{4K}}$$

Combinations of Springs

1. Series Combination

Ends of both spring are joined with each other.

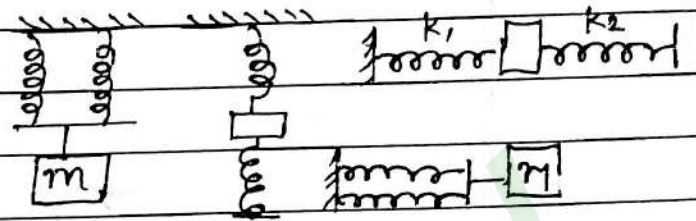


$$\frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow$$

$$k_e = \frac{k_1 k_2}{k_1 + k_2}$$

❖ Parallel Combination

One end is joined with Fixed end and other with mass.



$$k_e = k_1 + k_2$$

➤ Cutting of spring

$$k \propto \frac{1}{l}$$



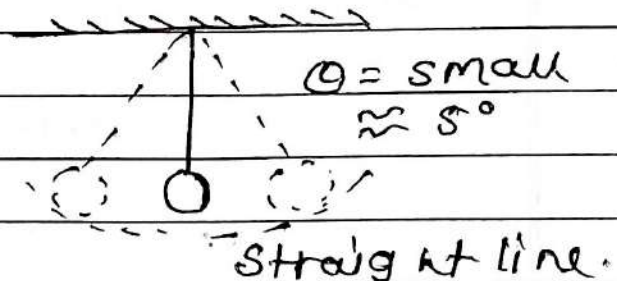
l/n l/n l/n

nk nk nk

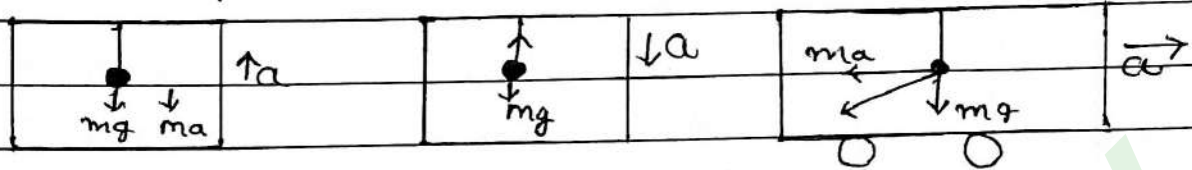
➤ Simple Pendulum for small oscillations

$$T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}}$$

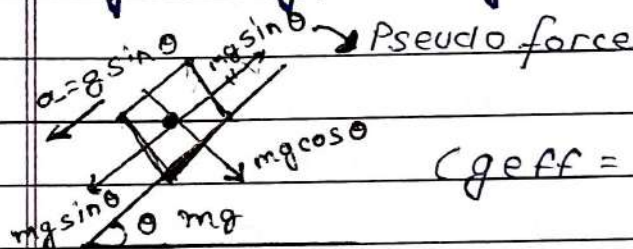
net acceleration



► Concept of $g_{\text{effective}}$



$$(g_{\text{eff}} = g + a) \quad (g_{\text{eff}} = g - a) \quad (g_{\text{eff}} = \sqrt{g^2 + a^2})$$

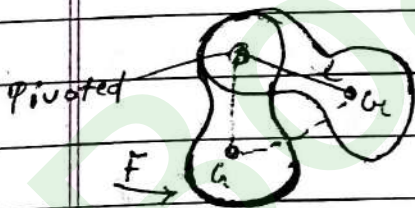


$$(g_{\text{eff}} = g \cos \theta)$$

► Angular SHM

$$\left\{ \begin{array}{l} \tau = -k\theta \\ T = 2\pi \sqrt{\frac{I}{k}} \end{array} \right.$$

► Physical Pendulum → Mass distributed



$$T = 2\pi \sqrt{\frac{I}{mgl}} \quad \leftarrow \begin{array}{l} \text{moment of inertia} \\ \text{about pivoted point} \end{array}$$

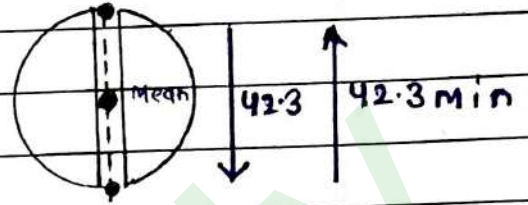
$l \rightarrow$ distance b/w centre of gravity & pivoted point.

S.H.M of a body in a tunnel along any chord of earth

① Along diameter of earth

S.H.M about centre of earth

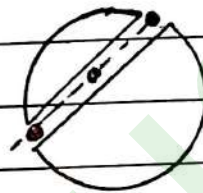
$$T = 2\pi \sqrt{\frac{R_e}{g}} \approx 84.6 \text{ mins}$$



② Any other chord

S.H.M about centre of chord

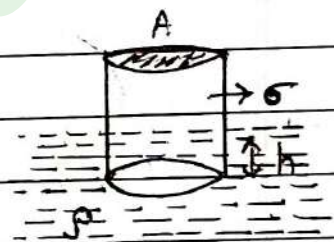
$$T = \text{Same} = 84.6 \text{ min}$$



Oscillation of floating body

Floating: equilibrium

slightly displace $\Rightarrow$ S.H.M



$$T = 2\pi \sqrt{\frac{m}{A\rho g}}$$

$m \rightarrow$ mass of body

$\rho \rightarrow$ density of fluid

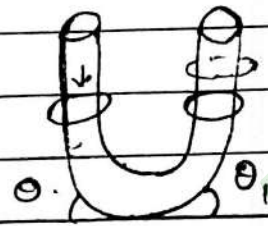
$A \rightarrow$ cross sectional Area.

$$T = 2\pi \sqrt{\frac{\sigma A L}{A\rho g}} = 2\pi \sqrt{\frac{\sigma L}{\rho g}}$$

$$T = 2\pi \sqrt{\frac{h}{g}}$$

► Oscillation of liquid column

$$T = 2\pi \sqrt{\frac{m}{A\rho g (\sin \theta_1 + \sin \theta_2)}}$$



For U-tube, $\theta_1 = \theta_2 = 90^\circ$

$$T = 2\pi \sqrt{\frac{m}{2A\rho g}}$$

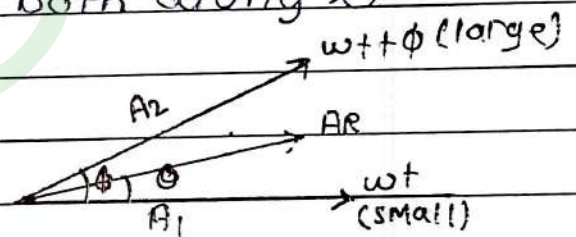
► Combination of two or More S.H.M

1 In same direction (both along x)

$$x_1 = A_1 \sin \omega t$$

$$x_2 = A_2 \sin (\omega t + \phi)$$

$$x_R = x_1 + x_2 \text{ (SHM)}$$



$$A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos \theta}$$

$$\tan \theta = \frac{A_2 \sin \phi}{A_1 + A_2 \cos \phi}$$

$$x_R = A_R \sin (\omega t + \phi)$$

(Resultant SHM)

Note - $\cos \omega t = \sin (\omega t + 90^\circ)$

$-\cos \omega t = \sin (\omega t - 90^\circ)$

⇒ Find the equation for resulting SHM

$$x_1 = 5 \sin (\omega t + \pi/6)$$

$$x_2 = 5 \cos \omega t$$

$$x_1 = 5 \sin (\omega t + 30^\circ) \rightarrow 60^\circ$$

$$x_2 = 5 \cos \omega t = 5 \sin (\omega t + 90^\circ)$$

$$x_R = 5\sqrt{3} \sin (\omega t + 60^\circ)$$

